M.E. CSE PROJECT REVIEW · MODULES 1 AND 2

Romanised Tamil queries break RAG retrieval and a 0.3 ms normaliser repairs most of it

Script-aware retrieval for Tanglish queries over a bilingual Tamil–English document base

STUDENT

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GUIDE

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REVIEW

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Module 1

Tanglish retrieval over bilingual documents is an open gap

Problem, motivation, literature, design

People ask in Tanglish while the answers sit in Tamil or English documents.

"scholarship ku last date enna?" A G.O. may exist only in Tamil and a circular only in English. RAG is tested in English.

One question arrives in four forms that share almost no characters

FORM	EXAMPLE QUERY
English	Who is responsible for the Israel-Hamas conflict
Tamil script	இஸ்ரேல் ஹமாஸ் மோதலுக்கு யார் பொறுப்பு
Romanised Tamil	Israel Hamas mothalukku yaar poruppu
Code-mixed	Israel Hamas conflict ku yaar responsible

ChatGPT reads Tanglish but still needs retrieval to find the right document

01

Thousands of G.O.s and circulars do not fit in one prompt

02

The retrieval step collapses on romanised Tamil: dense MRR@10 is 0.143

03

Translate-to-English pipelines never search Tamil-only documents

04

Paid APIs raise cost and privacy issues for citizen data

Six objectives take the study from baseline to public benchmark

OBJECTIVE	STATUS
O1 Four-form queries, Tamil + English docs	Started
O2 Measure the drop per form	Done
O3 Normalisation and its latency	Done
O4 Bilingual benchmark and chunking (M3)	Planned
O5 Router and answer quality (M4)	Planned
O6 Paper and release (M5)	Planned

Prior work covers Hindi or native Tamil but never Tanglish over bilingual documents

WORK	SETTING	WHAT IS MISSING
FIRE MSIR 2013–16	Hindi, Bengali IR	No Tamil retrieval
Chari et al. 2025	Latinised non-Indic	No Indic language
Amiraz et al. 2025	Arabic–English RAG	No Tamil; native script
Hira et al. 2026	Romanised Hindi, Bengali	Parallel docs; no Tamil
SchemeAI 2026	TN scheme chatbot	English index; no metrics

The proposed system adds the script handling SchemeAI lacks

Existing (SchemeAI-style)

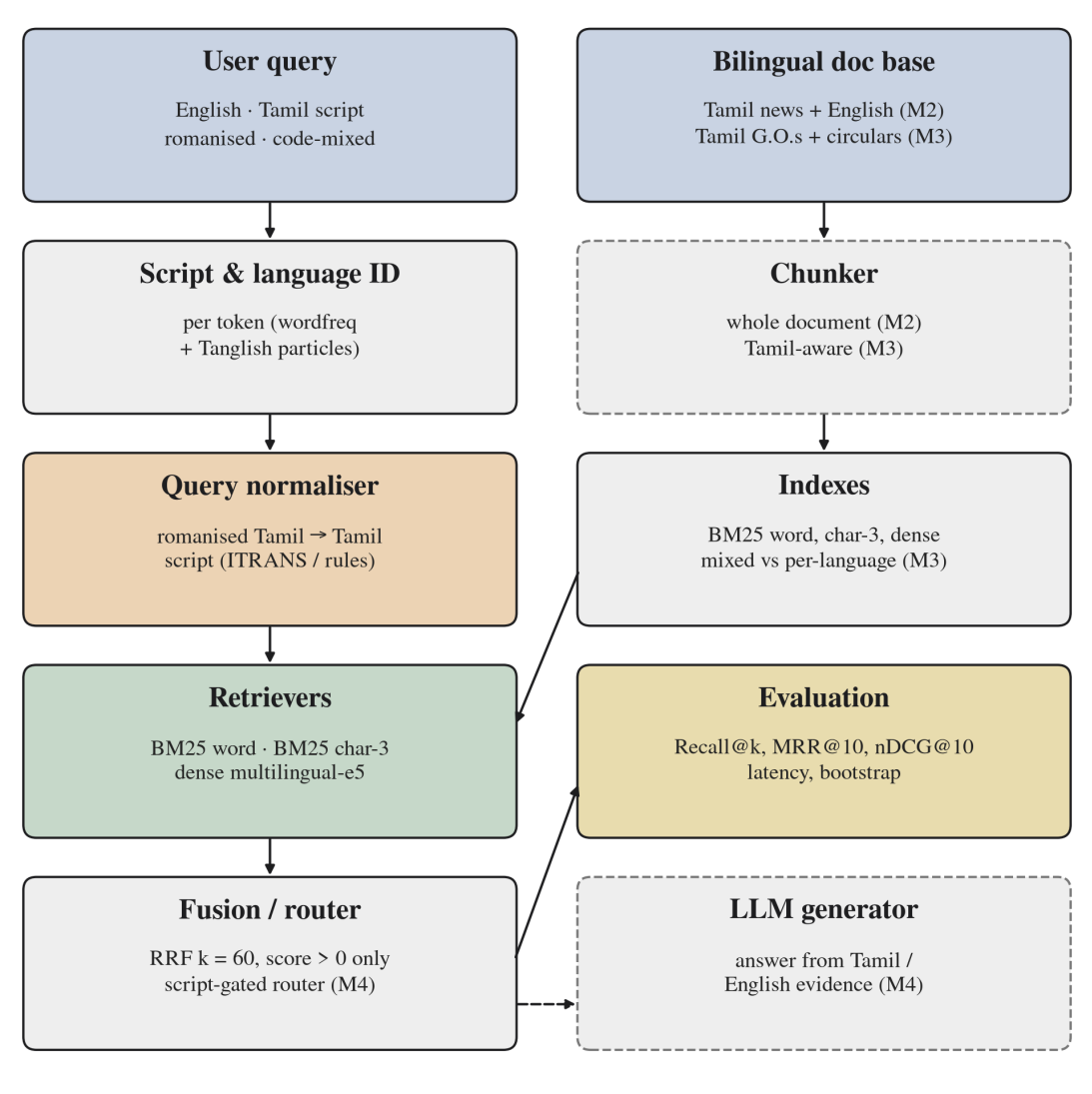
English-only index. Query used as typed. English MiniLM embedder. Output translated to Tamil. No retrieval metrics.

Proposed

Bilingual Tamil + English base. Token language ID and transliteration. Char 3-gram BM25 + multilingual dense. Fusion measured per form.

The query is normalised before three retrievers and rank fusion

Solid boxes are built in Module 2. Dashed boxes are the chunker, router and generator in Modules 3 and 4.



Two modules are complete and three carry the thesis to May 2027

M1 · Done

Problem, survey, gap and architecture

M2 · Done

Data, normaliser, five retrievers and results

M3 · Dec–Jan

Bilingual human Tanglish benchmark and chunking

M4 · Feb–Mar

Router, generation and answer evaluation

M5 · Apr–May

Paper, public release and thesis

Module 2

Eighty-four retrieval runs on a 2-core CPU

500 Tamil news documents with English translations · 2,154 queries per form · 8 query sets · 5 retrievers

Every information need is tested in all four forms

500

Tamil documents

2,154

Queries per form

8

Query sets

180,936

Ranked lists

Normaliser and character n-grams target the script mismatch

Normaliser

Token language ID keeps English words. Romanised Tamil goes to Tamil script by ITRANS or Tanglish rules.

Retrievers

Word BM25, char 3-gram BM25, mE5-small dense, RRF and RRF3 hybrids.

Evaluation

Recall@k, MRR@10, nDCG@10, latency, query and cluster bootstrap.

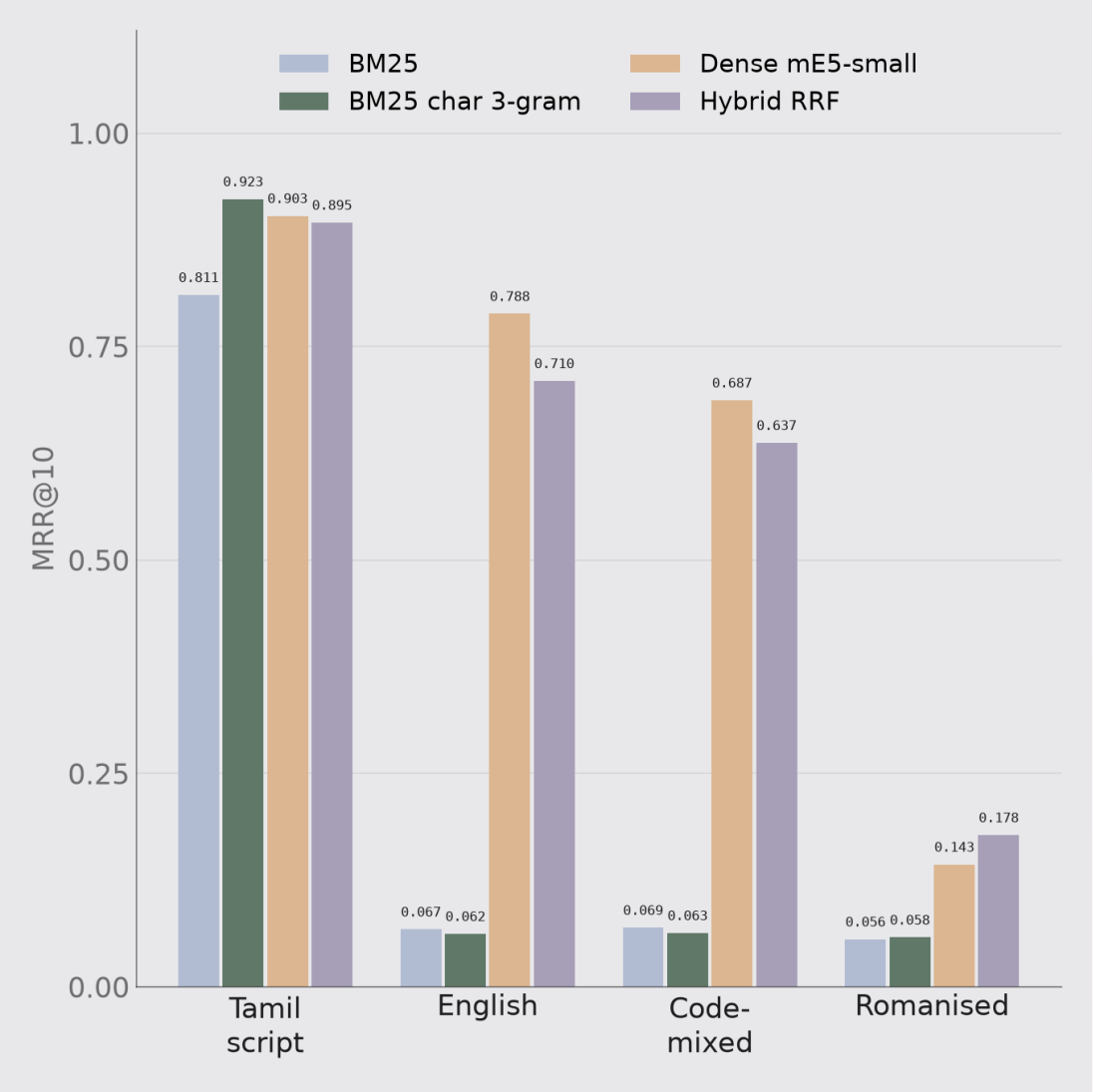
Results

Romanised queries fail and transliteration fixes most of it

Main results, normalisation, fusion and examples

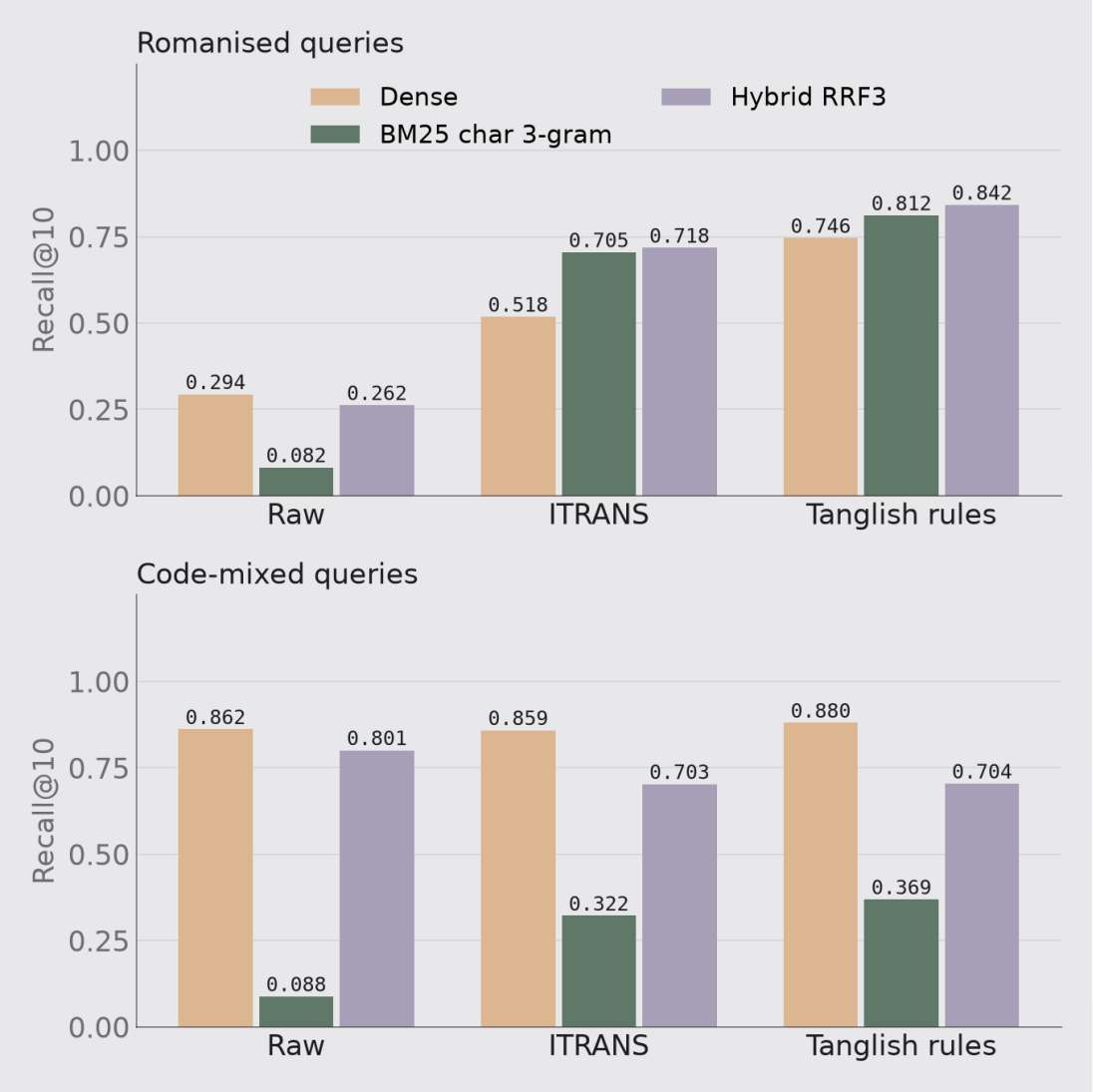
Romanised queries fall below MRR 0.18 while Tamil script reaches 0.92

Dense Recall@10 for romanised queries is 0.294. Seven in ten questions miss the top ten.



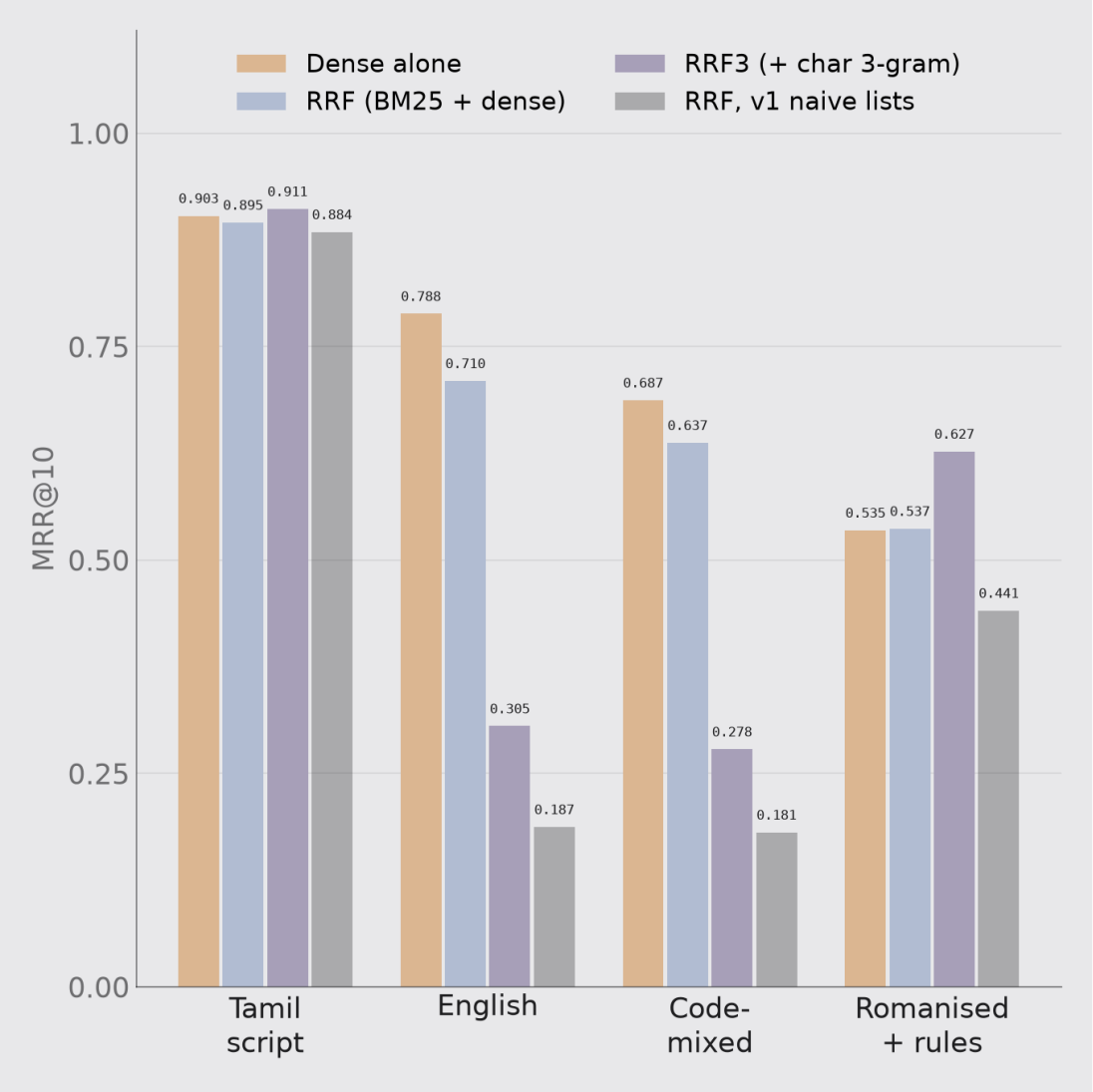
Tanglish rules lift romanised Recall@10 from 0.08 to 0.81

Rules + char 3-gram BM25: MRR 0.058 → 0.684 ($p < 0.001$) for 0.34 ms per query. Rules beat ITRANS everywhere.



Equal-weight fusion cuts English-query MRR from 0.79 to 0.71

BM25 matches nothing across scripts yet RRF trusts it. Gate fusion by script and drop zero-score lists.



Transliteration rescues real queries but misses unmarked long vowels

QUERY GIVEN TO THE DENSE RETRIEVER	GOLD RANK
thakkudhalil Colombia raanuva thiraigal uyirizhappu	not in top 10
தக்குதலில் Colombia ரானுவ திரைகல் உயிரிழப்பு (rules)	1
Vadakku maagaana paadasaalai seerudai neelam korikkai	not in top 10
வடக்கு மாகான பாடசாலை சீருடை நீலம் கொரிக்கை (rules)	1
Muthalaq thadai masoda Rajyasabha	not in top 10
முதலக் தடை மசொட ரஜ்யசப (rules)	not in top 10

The dataset's code-mixed queries are mostly English

22%

No Tamil word

13%

Identical to English

1.7

Tamil tokens per query

Four findings close Module 2

01

Romanised Tamil breaks retrieval: dense MRR@10 0.143 against 0.923 for Tamil script

02

Tanglish rules + char BM25 recover most of it: 0.684 at 0.34 ms

03

Equal-weight fusion hurts across scripts: 0.788 → 0.710

04

Public code-mixed queries are English-heavy, so human data is needed

Closest references anchor the claim and the baselines

- Lewis et al. NeurIPS 2020: retrieval-augmented generation
- Cormack et al. SIGIR 2009: reciprocal rank fusion
- Sequiera et al. FIRE 2015: mixed-script IR overview
- Chari, Ounis, MacAvaney SIGIR 2025: script gap in neural IR
- Amiraz et al. ArabicNLP 2025: cross-lingual cost in bilingual RAG
- Hira, Mire, Sap EMNLP 2026: language of evidence in code-switched RAG
- Rekha et al. IJARSCT 2026: SchemeAI scheme chatbot
- nlpctx 2026: tamil-english-retrieval dataset

NEXT

Module 3 tests a bilingual human Tanglish benchmark and the mixed-index bias

About 500 human questions over Tamil G.O.s and English circulars, each tagged with its gold-document language.

DEC–JAN

Benchmark, bias test, chunker, IndicXlit and SchemeAI baselines

FEB–MAR

Script-gated router and answer evaluation

APR–MAY

Paper, thesis and public release